

A Brief Note on Dynamic Antisymmetry¹, Merge-based Theory², and its Implications to Early Child English Possessive {‘s} and the Setting of Word Order.

Joseph Galasso, California State University, Northridge~Linguistics

In recent personal correspondences both with Andrea Moro ([IUSS](#)) (regarding Basal Ganglia Grammar), and Elena Babatsouli (University of Louisiana at Lafayette), (regarding a new entry proposal for ‘Oxford Bibliographies in Linguistics’, cf. Essential Readings in ‘Early Child Acquisition of Possessive’³), the question addressed to me was whether I thought a ‘new twist’ could be crafted out of general conclusions reached in some of my previous research on the acquisition of functional categories⁴. Thus, the writing of this brief note. *I can suggest that a previously un(der)-reported correlation may hold between the onset of early child English possessive {‘s} and the fixing of word order.* Merge is inextricably linked to Dynamic Antisymmetry (DA) insofar that any exclusive ‘merge-only’ stage would only support a strictly *symmetric/unordered* SET{ α, β } (where higher functional features would not project and where word order would remain variable). Antisymmetry (non-mirror structures), leading to the formulation of hierarchical fixed word order, as expressed in Moror’s work on DA⁵, can only come about via *displacement* properties having to do with *recursive* MOVE, projecting a subsequent structure-dependent PAIR< α, β > whilst dispensing with the former structure-independent SET{ α, β }. This shift from SET to PAIR can only arise via MOVE.

$(\text{SET}\{\alpha, \beta\} \rightarrow \text{PAIR} \langle \alpha, \langle \alpha, \beta \rangle \rangle).$

Any first-order exclusive MERGE stage would not be able to support displacement operations—hence, exclusive MERGE prevents projection of all hierarchical structures. The argument herein claims that since there is only evidence for initial SET ‘sister-sister’ relations⁶ as attested in my stage-1 data—calling for an exclusive merge-based theory of child language (fn.4), whereby sister relations subserve symmetry (and where non-hierarchical structures persist)—the implication is that MERGE/SET not only effectuates a delayed onset of inflectional Possessive {‘s}, but also a delayed setting of correct Word Order. Both linguistic phenomena are the result of MOVE, a DA result.

¹ (Paper submitted to <lingbuzz.net> December 2024). See works by Andrea Moro on antisymmetry (Institute for Advanced Study [IUSS](#)), For a brief overview of Moro’s DA as highlighted in Chomsky’s 2013 *Lingua* paper, see Galasso <https://ling.auf.net/lingbuzz/007267>

² See works by Joseph Galasso on a merge-based theory of child language acquisition. (For review of Acquisition of Possessives, see Radford & Galasso (1998). <https://www.csun.edu/~galasso/arjg.pdf>

³ <https://www.oxfordbibliographies.com/display/document/obo-9780199772810/obo-9780199772810-0292.xml>

⁴ https://www.academia.edu/43939141/The_Acquisition_of_Functional_Categories_A_Case_Study

⁵ See Moro file:///C:/Users/csuni/Downloads/moroEtAl_20_Unstable-stru.3.pdf

⁶ See Galasso, ‘ABABABA grammars’ file:///C:/Users/csuni/Downloads/galasso_22_The-Recurrent-.pdf

As a brief update to my earlier work on early child syntax (see fn2 for review), I'd like to suggest that early child possessive structures can find more current theoretical expansions by correlating its projection to both (i) fixed child word order, and (ii) the full range of onsets of inflectional morphology to (iii) the displacement operation MOVE—where the cited genitive/possessive structures are but one manifestation. More current analyses might be made in support of Andrea Moro's 'Dynamic Antisymmetry'(DA) model which, *inter alia*, suggests that in order for a given INFL phrase to be labelled and therefore ordered, (thus providing for a H(ead) of P(hrase)), hierarchical movement must ensue, thus breaking with flat-symmetric sisterhood relations (e.g., an unordered SET{ α , β }). My 2001 analysis on the nonexistence of possessive ('s) structures, as correlated with early child variable word order, I feel, can be easily extended to capture Moro's premise that any putative 'Merge-only' stage, as found in my early stage-1, would not entail the hierarchical capacity to support the correct setting of the full range of inflectional morphology, upon which the correct setting word order depends, as attested in these data.<> See Ref. Paper. Goodluck H, Kazanina N. (2020) 'Fragments Along the Way: Minimalism as an Account of Some Stages in First Language Acquisition'. PMID: 32457672. More recent research has shown that in early stages word order can be variable: strings that must be interpreted as Subject—Verb, Verb—Subject, Object—Verb and Verb—Object are attested in languages with SVO order (Tsimpli, 1992 [quoted in Galasso, 2001], Galasso, 2001). Thus, it may be the case that at a very early stage the child combines two words without attention to headedness. Nonetheless, the evidence favors the rapid development of a system in which asymmetric Merge is found in child language.<> See paper 'Note on 'Problems of Projection' (Chomsky's 2013 *Lingua* paper): <https://ling.auf.net/lingbuzz/007267>

Theory: DA. Early child stage-1 'merge-based' structures make licit only simpleton sister relations {x, y} which project prosaic lexical items at the expense of a fully functional/categorical stage showing *inter alia* Possessives/Genitives {'s}. Furthermore, in addition to the classic benchmarks discussed in the literature regarding under/non—specification of INFL/Agreement, I feel a new extension can be made towards Moro's idea—supported by my data—that DA (viz., the lack of recursive hierarchy as found in simpleton merge-operations), offers us a window into how the eventual setting of fixed word order in child syntax is connected to the onset of Possessive structures. I suppose this might constitute a new 'twist' in how the Minimalist Program can capture, as a universal operation, a language-specific (English) fixation of word-order to the projection of possessive {'s}.

Data (Early DP-Projections). A further observation has to do with the fact that word order errors seem also to correspond with the absence of feature specification associated with the Head of DP: (possessive 'S). It is observed in the data that the first attempts to project a Possessive construction (DP) (Type-A/within a non-head) fail to realize the target order. (NB. We say 'non-head' because within simpleton Merge, both *symmetric* nouns [N, N] could equally project as a viable head. (Recall, it's the specification of a single head, as pulled from the lexical array, (a head that then raises to a higher/functional projection), which allows extension of both Spec and Comp, yielding:

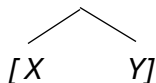
[Move<[Merge]], an extension of previous X-bar Theory [Spec, [Head, Comp]]).

Consider the following examples below (1), where the Head Possessive 'S is unspecified:

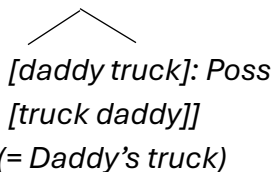
The template for such movement, as we discuss throughout this paper, requires a mother-daughter/hierarchical tree diagram, as drawn below:

(1) Type-A/MERGE

a. Sister-sister {X, Y}



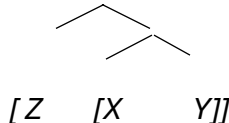
a. Sister-sister {X, Y}



Type-B/MOVE Phrasal Word

Order/Poss(essive)

b. Mother-daughter {Z {X, Y}}



b. Mother-daughter {Z {X, Y}}

[daddy's [daddy truck]]⁷ : Poss

Simpleton Merge symmetry where either Noun can project as Head suggest a bricolage Merge independent of structure.

(2) Possessive {'s} AGE/ OCCURRENCE IN OBLIGATORY CONTEXTS AGE

Age:	2;3-3;1	0/118 (0%)	'Merge' Stage-1
	3;2-3;6	14/60 (23%)	'Move' Stage-2

(3) Lack of Possessive {'s}: MERGE/Stage-1: Ages 2;3-3;1. (note mixed word order).

- | | |
|------------------|--------------------|
| a. daddy ∅ truck | d. bottle ∅ baby |
| b. baby ∅ car | e. key ∅ mama |
| c. baby ∅ bottle | f. hair ∅ me (=my) |

⁷ Note it's the movement/raising of the lexical item (creating 'mother-daughter' hierarchy) which enables the phrase to be labelled (as a DP or VP, etc.). Examples such as [boat-house] vs [house-boat] (the former a kind of 'house', the latter a kind of 'boat', pose ambiguous problems for young children regarding interpretation. In order to disambiguate, raising of one of the items must ensure and the structure must project as [house [house-boat]] (= a kind of boat).

For some of my fuller discussions, see paper: 'Why Move?', see <https://osf.io/5he2x>

(4) Onset of Possessive {'s}: MOVE/Stage-2: Ages 3;2-3;6.

a. Baby's hair (file: 25) d. It's [Zoe's bike] (diary)

b. Car's home (file: 25) e. Where is [Zoe's bottle?] (diary)

c. Dad's keys (file: 25) f. A boy's bed (diary)

NB. We note in the data that once the Possessive {'s} projects as found in (4), all DP-structures project correct word order. There are no examples found in the data of e.g., [*bottle baby's] (cf.3d).

Word Order: Exclusive Merge-stage: Token counts of VS.

(5)

MERGE

VP

/ \

V Spec

| |

| |

a. run Baby

b. cook Daddy

c. work Bike

d. go Plane

Spec V

a'. Baby run

b'. Daddy cook

Table 5.1 (Stage-1 Merge)

Token counts	SV	VS	SVO	Other (xyz)
(Stage-1 files): n.	87	78	290	15

Mirror image projections: *V, Spec (VS) // Spec V (SV). SET{x, y // y, x}. This extends to Possessive {'s} as seen as triggering correct Spec-Head. (cf. (1), Type B).

Objects. Examples of OV/VO structures are also found throughout the range of the data. Word order errors seem to manifest in our Merge-based stage-1, ending when MOVE is fully operational (which we delimit as stage-2). Token examples are presented below in (6):

(6) VO

a. kick the dog (2;0)

b. want my car (2;4)

c. no cut train (2;8)

d. cook pasta (2;9)

(=Dad cooks pasta)

e. want bottle (2;4)

(=Baby wants bottle)

f. make a house (3;3) (= He makes a house)

OV

h. dog kick (1;11)

i. baby kick (2;5)

j. ball a kick (2;8)

k. a egg cook (2;11)

l. a cookie eat (3;0)

m. No baby hit (3;2)

(=I don't hit the baby)

(7) OV within Merge

Table 7.1 (Nom Case correlates to correct SV order)

VP			
/	\	Nom Case	Default Acc Case
Comp	V	SV: (n.418)	128
		VS: (n.151)	0
			24
h.' dog kick			
k.' a egg cook			
l.' a cookie eat			

SV/VS Token Examples

SVx: I play, He cut, He do it.

VS: eat Me, run Baby, Kick Baby,

open Me, help Me, work Bike

Notes.

(Paper submitted to <lingbuzz.net> December 2024). See works by Andrea Moro on antisymmetry (Institute for Advanced Study [IUSS](#)), For a brief overview of Moro's DA as highlighted in Chomsky's 2013 *Lingua* paper, see Galasso <https://ling.auf.net/lingbuzz/007267>

See works by Joseph Galasso on a merge-based theory of child language acquisition. (For review of Acquisition of Possessives, see Radford & Galasso (1998). <https://www.csun.edu/~galasso/arjg.pdf> <https://www.oxfordbibliographies.com/display/document/obo-9780199772810/obo-9780199772810-0292.xml>

https://www.academia.edu/43939141/The_Acquisition_of_Functional_Categories_A_Case_Study

See Moro file:///C:/Users/csunl/Downloads/moroEtAl_20_Unstable-stru.3.pdf

See Galasso, 'ABABABA grammars' file:///C:/Users/csunl/Downloads/galasso_22_The-Recurrent-.pdf

Note it's the movement/raising of the lexical item (creating 'mother-daughter' hierarchy) which enables the phrase to be labelled (as a DP or VP, etc.). Examples such as [boat-house] vs [house-boat] (the former a kind of 'house', the latter a kind of 'boat', pose ambiguous problems for young children regarding interpretation. In order to disambiguate, raising of one of the items must ensure and the structure must project as

[house [~~house~~-boat]] (= a kind of boat).

For some of my fuller discussions, see paper 'Why Move?' <https://osf.io/5he2x>